

Name: ___Gabrielle D. Tez, Ethan R. Gabrielle S. ___ Date: _____

Class: _____

MyDesign Portfolio

Element G

Element G: Construction of a Testable Prototype

Final Prototype Iteration Explanation (G1)

The Irrigation system was able to drip water to a 12x12 area, however there needs to be more balance and have a level ground/slope to allow for an equal and more efficient system. Other than that the system worked great for the criteria that was presented to us.

Date: 3/14 - 3/28

Steps	Details	Tools and Manufacturing Methods used
Step 1 3/14	Collected and sorted materials	Boxcutter 
Step 2 3/18	1. Measured PVC 2. Cut PVC 3. Drilled holes for water to come out	Drill, drill bit, measuring tape, mallet, sharpy, PVC saw 

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Step 3 3/20	1. Finished drilling the holes in the pipe 2. Sanded the edges of the pipe that was cut 3. Put whole system together with fittings	Drill, sand paper, PVC saw 
Step 4 3/24	Tested	We need to add elevation to the start to let the water be able to run to the bottom and outer edges of the system (saw, cardboard, duct tape) 
Step 5 3/28	Final Testing Prototype	Need to add leveling and balance to the added height 

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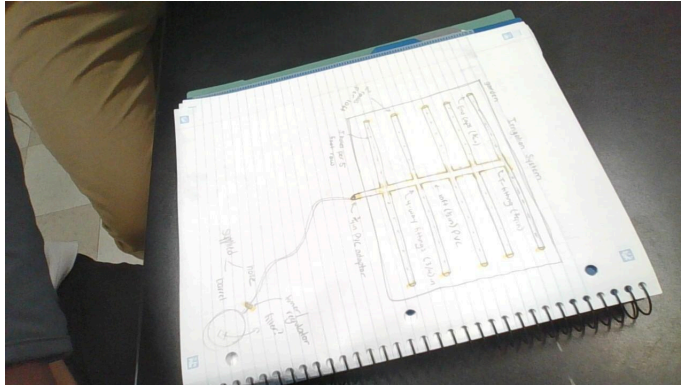
Material List

Material name	Description	Where material was purchased	Cost per unit	# units	Total cost per item
1/2 in PVC Pipe	10ft	Home Depot	\$4.71	6	\$28.26
1/2 in PVC T fitting	fitting	Home Depot	\$0.82	1	\$0.82
1/2 in PVC cross fitting	fitting	Home Depot	\$5.21	4	\$20.84
1/2 in PVC female adaptor	Adapter from hose to PVC	Home Depot	\$2.98	1	\$2.98
1/2 in PVC caps	End piece for PVC	Home Depot	\$0.82	10	\$8.20
Timer/regulator	Water amount technology	Home Depot	\$19.94	1	\$19.94
Total Cost				23	\$81.04

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Sketch of System



Construction of the Final Prototype (G2)

Design Requirement	How will this prototype allow you to collect objective, measurable data?
Irrigate a 12x12 area	PVC piping was long enough to water the entire area for every plant to be watered.
Last 10 days	Timer that allows for controlled amounts of water that will allow a full barrel of water to be evenly distributed throughout the day.
Use collected water	Water is brought from the roof and gutter system and then collected in the barrel.
Portable	The system is able to be removed and picked up.
Sustainable/Efficient	Met all of the criteria

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Testing or Modeling of Attributes (G3, G4) and Addressing Non-Testable Attributes (G4)

Attribute	Description	Describe your prototype's functionality to test this attribute
Timer/sensor	A device that regulates the time and the amount of water that is let out from the bucket to the pipes.	Our design passed
Water Hose	The water hose is connected to the pressure connecting to the PVC pipes.	This was a given element
PVC pipes	PVC material pipes that will lay in the rows of the garden. It allows the water to flow from the water hose to plants to flow into a drip irrigation system irrigating the plants	Our design passed with water being disturbed throughout the pipes to the plants covering the entire 12x12 area.
Drip System	Holes in the PVC allow water to drip out. The holes in the PVC allows the drip system of water to flow onto the area on the ground around the plants.	Our system passed, but needs a balancer to be able to equally distribute the water more.
Grate	A frame of metal bars, layed on the top water tank covering the hole. Catches leaves and other debris from entering the barrel.	Did not apply to test



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Barrel	Bucket that holds the water. The bucket holds the water that will flow to the PVC pipes.	The product was given to us.
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Addressing Non-Testable Attributes (G4)

Attribute	Justification for why this attribute cannot be modeled or tested with your prototype
Above Ground	Because you cannot test if it is above ground it is common knowledge.

Testing Plan

Testing Plan Details

	requirement	testing setup and procedure	test cases and number of trials	expected range of results	pass/fail criteria, goal
1.	Irrigate a 12 x 12 plot of land	Set up the irrigation system. Observe as the water flows to the plot of land. Measure the amount of water that goes to each sq ft.	1 trial	The 12 x 12 plot of land would be evenly irrigated. Each plant would receive enough water to grow properly. Each plant should receive	Fail: The plants do not receive the proper amount of water. Pass: All plants receive enough water.



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				one and a half gallons of water over 10 days.	Goal: Irrigate a 12 x 12 plot of land evenly
2.	Reliable for 10 days	The irrigation system would be able to independently support a 12x12 plot of land of plants for at least 10 days. Being able to cycle water and collect water to support plants.	1 Trial	We would expect the owner to have a 125 gallon reservation.	Fail: Not provide enough water for 10 days. Pass: Provide enough water for 10 days. Goal: Provide enough water for 11 days.
3.	Amount of water produced per day	Set up the irrigation system and let the water run for a certain amount of time. Place the system on concrete or a place where the amount of water let out of each row is visible to make sure that each plant is being watered equally. Then make sure that there is enough water to be able to water the plants every day for up to ten days while the plants are still receiving the proper amount.	1 trial	The plants are water sufficiently with equal and healthy amount of water for up to ten days.	Fail: There is either too much or too little amount of water that is let out from the system. It overwater or underwater plants, or it will run out of water before the client returns from vacation. Pass: The plants are watered and the water in the tank is able to last for 10 days Goal: Plants are perfectly watered, water is evenly distributed, and

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					the water in tank is able to hold and distribute enough water for 10 days
4.	Ability to collect natural water	Set the tank up and connect it to the downspout with a filter that is in between to catch debris. Let it rain and see how much water the tank collected and how clean the water is without the help of municipal water.	1 trial	The tank collects water from the runoff from the roof through the down spout, filters out debris that have landed in the gutter/roof, and also holds the water for 10 days without using municipal water.	Fail: Cannot collect enough water to sustain the system for 10 days. Pass: Can collect enough water to sustain the system for 10 days. Goal: Sustain the system for 11 days.
5.	Portable	Set a timer for 10 minutes. Attempt to disassemble the system and effectively move it.	1 Trial	1. The System fails to be taken apart effectively and cannot be moved. 2. The system is taken apart effectively and can	Fail: You can't move it without destroying or changing the structure Pass: It can be effectively disassembled and moved easily Goal: To be able to be disassembled in under 10 minutes and be

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				be moved within 10 minutes.	easily moved.
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